

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 16: Semiconductors, Logic Gates & ARINC 429 Buses

Syllabus Source: CAE Oxford (pp. 167-200 (Elec)) | Pass Target: >= 90% AviationExam

1. SOLID-STATE SEMICONDUCTOR DEVICES

Solid-state electronics form the core of aircraft digital control units (FADEC, FMS, Fly-by-Wire FCCs, EGPWS):

- PN Junction Diode: Allows current to flow in one direction only (forward bias); blocks current in reverse. Used for rectification and reverse current protection.
- Zener Diode: Operates in reverse breakdown mode at an exact, predetermined voltage (Zener voltage). Used as high-precision VOLTAGE REGULATORS and voltage reference standards.
- Transistors (BJT / MOSFET): Three-terminal semiconductor devices (Base, Collector, Emitter or Gate, Drain, Source). Used as electronic solid-state switches and amplifiers.

2. DIGITAL LOGIC GATES: TRUTH TABLES & AERONAUTICAL LOGIC

Flight control computers and warning systems utilize Boolean logic gates to evaluate flight parameters:

Gate Type	Boolean Equation	Output Condition & Aircraft System Application
AND Gate	Output = A . B	Output is HIGH (1) ONLY IF ALL inputs are HIGH (1). Example: Cabin altitude horn = Cabin Alt > 10,000 ft AND Aircraft Airborne.
OR Gate	Output = A + B	Output is HIGH (1) IF ANY input is HIGH (1). Example: Master Caution = Low Hyd Press OR Low Fuel Press OR Gen Trip.
NOT Gate (Inverter)	Output = NOT A	Output is the exact opposite of input (1 becomes 0; 0 becomes 1).
NAND Gate	Output = NOT (A . B)	Output is LOW (0) ONLY if all inputs are HIGH (1). Inverted AND gate.
NOR Gate	Output = NOT (A + B)	Output is HIGH (1) ONLY if all inputs are LOW (0). Inverted OR gate.

3. AVIONICS DIGITAL DATA BUSES: ARINC 429 STANDARD

Modern avionics units (LRUs) communicate digitally over standardized serial data buses rather than individual heavy point-to-point analog wiring harnesses:

- ARINC 429 Architecture: The universal commercial standard. It is a SIMPLEX (one-directional) data bus. Transmits data from one single transmitter to up to 20 receivers across a twisted shielded wire pair.
- Word Structure (32 Bits): Every ARINC 429 transmission is an uncompressed 32-bit digital word: Bits 1-8: Label (identifies parameter, e.g. 203 = Pressure Altitude, 310 = Present Position Latitude); Bits 9-10: SDI (Source/Destination Identifier); Bits 11-29: Data Field (BCD or BNR numeric value); Bits 30-31: SSM (Sign/Status Matrix: Normal, Test, Failure Warning); Bit 32: Parity Bit (Odd parity standard to detect transmission errors).
- Transmission Speeds: Low-speed mode: 12.5 to 14.5 kbps; High-speed mode: 100 kbps.

- Advanced Aircraft Buses: Boeing 777 uses ARINC 629 (bidirectional, multi-transmitter); Airbus A380 / A350 and Boeing 787 use AFDX (Avionics Full-Duplex Switched Ethernet, 100 Mbps).

[!] EXAM TRAP (AVIATIONEXAM) - ARINC 429 Exam Traps Summary

- Simplex Bus: Data flows in ONE DIRECTION ONLY! (Two-way communication requires two separate wire pairs). - Word Length: Exactly 32 bits! - Parity: Bit 32 is always ODD parity. - Physical Wire: Twisted shielded pair with 78-ohm characteristic impedance.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): ARINC 429 Data Word Decoding**

Architecture: Simplex point-to-multipoint, 1 transmitter to up to 20 receivers, 32 bits per word. Word Breakdown: - Bits 1-8: Label (parameter ID, e.g. 204 for Altitude, in reverse octal). - Bits 9-10: SDI (Source/Destination Identifier). - Bits 11-29: Data Field (Binary or BCD engineering units). - Bits 30-31: SSM (Sign/Status Matrix: Normal, Test, Failure, or +/- sign). - Bit 32: Parity Bit (Odd parity for error detection).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the bit structure length of a standard ARINC 429 digital avionics data bus word?**

- [A]: 16 bits
- [B]: 32 bits
- [C]: 64 bits
- [D]: 128 bits

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The standard ARINC 429 data word consists of exactly 32 bits, comprising an 8-bit label, 2-bit SDI, 19-bit data field, 2-bit SSM, and a 1-bit parity check.

Q2: Which type of semiconductor diode is specifically engineered to operate in reverse breakdown mode to provide a stable reference voltage?

- [A]: Photodiode
- [B]: Zener diode
- [C]: Light Emitting Diode (LED)
- [D]: Germanium power diode

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Zener diodes are designed to conduct in reverse once a specific breakdown voltage is reached, maintaining a stable reference voltage regardless of current variations.

Q3: How many transmitters can be connected to a single ARINC 429 digital data bus wire pair?

- [A]: Only ONE transmitter (simplex operation)
- [B]: Up to 20 transmitters
- [C]: Unlimited transmitters
- [D]: Exactly two transmitters in full-duplex

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. ARINC 429 is strictly point-to-point or point-to-multipoint SIMPLEX: it permits ONLY ONE single transmitter per bus line, driving up to 20 listening receivers.